Cyber Security Lab

Course Code: 22CS11S5

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Course Outcomes: At the end of the Course, the Student will be able to:

CO1: Apply various cyber attack techniques and analyze their impact on systems and networks.
CO2: Develop and implement secure coding practices using cryptographic techniques and password security mechanisms.
CO3: Analyze and demonstrate common network attacks and defensive mechanisms through practical tools.
CO4: Apply appropriate cryptographic techniques and tools for ensuring secure data transmission and storage.
CO5: Evaluate and implement security configurations in various platforms including operating systems, browsers, mobile applications, and firewalls.

(Any 12 experiments of the following to be performed)

Implement the following techniques/programs

1. Implement the following Attacks: a) DictionaryAttack b) Brute Force Attack.
2. Write a program for static malware analysis that performs file signature checking, string extraction, and entropy analysis to detect suspicious files.
3. Write a program that reads an email body (as a string) and flags it as spam if it contains typical spam keywords like "lottery", "free", "click here", etc.
4. Write a program to
   a) Check the strength of a password based on uppercase, lowercase, digits, special characters, and length.
   b) Hash the password using bcrypt and save it to a file.
   c) Create a program to verify a login password by comparing it with the stored hash.
5. Write a script to interact with a security-related API (e.g., VirusTotal) to retrieve threat intelligence or breach data.
6. Write a program to extract hyperlinks from a given website using web scraping. Modify the program to detect suspicious or phishing-related URLs for cybersecurity analysis.

Perform the following experiments using relevant Tools

7. Demonstrate penetration testing tasks like port scanning, vulnerability scanning with Nmap tool.
8. Installation of Network traffic monitoring tools (Wireshark/tcpdump/Ethereal) and observe data transferred in client server communication using UDP/TCP and identify the UDP/TCP datagram.
9. Analyze Traffic Sniffing using ARP Poisoning.
10. Demonstrate DoS (Denial of Service) and DDoS (Distributed Denial of Service) attacks.
11. Demonstrate the following use cases of snort
    i. IDS
    ii. Perform real time traffic analysis and data packet logging
12. Demonstrate how to implement secure data storage, secure data transmission and create digital signatures
13. Demonstrate SQL Injection, Cross-Site Scripting (XSS) and File Upload vulnerabilities in web applications.
14. Illustrate the Phishing Attacks, Email Spoofing and Malware via Attachments vulnerabilities of E-Mail Application.
15. Analyze various wireless network features & components of any one of the Mobile Security Apps.
16. Examine the features of firewall in providing network security and set Firewall Security in windows.
17. Explore the steps to ensure security of any one web browser (Mozilla Firefox/Google Chrome).
18. Simulate Reconnaissance, Social Engineering, and Initial Access in an Advanced Persistent Threat (APT) Scenario.

References:

1. Dr Kutub Thakur , Dr Al-Sakib Khan Pathan, "*Cybersecurity Fundamentals: A Real-World Perspective"*, CRC Press, Taylor & Francis Group,1st Edition ,2020.
2. https://www.coursera.org/learn/foundations-of-cybersecurity
3. https://www.cybrary.it/